

WASP-43b

R-2691

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130311 · created 2026-08-31T01:30:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456362.67863 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.016
Transit depth (Rp/Rs)^2	2.53 +/- 0.51 [%]
Orbital Inclination (inc)	82.24 +/- 1.5
Airmass coefficient 1 (a1)	0.961 +/- 0.011
Airmass coefficient 2 (a2)	0.0224 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#2 - [76, 288]
Optimal Aperture	4.87
Optimal Annulus	6.76
Transit Duration (day)	0.0507 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.159 $\pm$ 0.016 vs 0.1594 (deviation 0.02 σ)
Duration fit vs published	0.0507 d vs 0.0512 d (deviation 0.09 σ)
Mid-time O-C	-1.5 min
Depth SNR	8.4
Residual scatter	1.13 %

Light curve

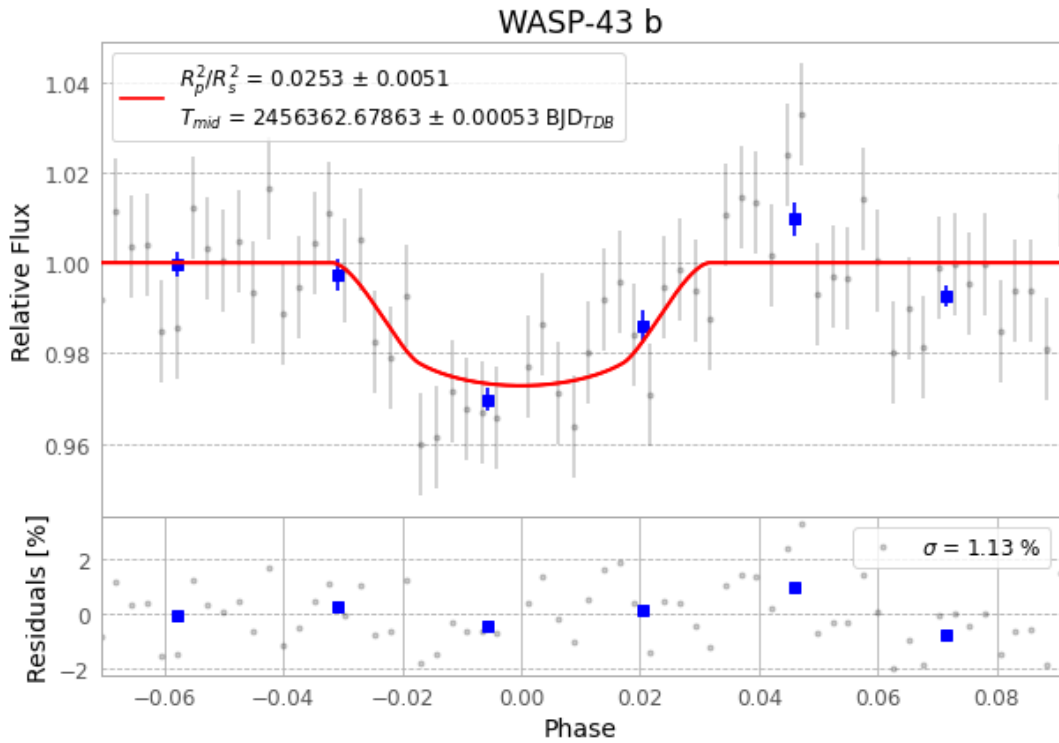


Figure 1 — FinalLightCurve_WASP-43 b_2013-03-10.png (SHA-256 bd8126124ffb5c04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [206, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 79941ef774f0a2b0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43130311024512.FITS → WASP-43130311055412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130311.log (8f26d00fac8...), FinalLightCurve_WASP-43 b_2013-03-10.png (bd8126124ffb...), FinalLightCurve_WASP-43 b_2013-03-10.csv (7abd6c6ebad0...)

Compute resources

Wall time 148.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 01:30Z.